

EMMANUEL KIBET KORIR

Cybersecurity Engineer | Full-Stack Developer | AI/ML

Email: ekorir555@gmail.com | GitHub: github.com/Korir555 | Portfolio: korir555.github.io/cybersecurity-portfolio

Nairobi, Kenya | Open to Remote and Freelance Opportunities

PROFESSIONAL SUMMARY

Cybersecurity engineer with expertise in penetration testing, threat detection, cryptography, and full-stack security tooling. Demonstrated ability to design and deploy production-grade security applications. Strong background in Python, Flask, React, machine learning, and secure coding practices. Building 11+ professional cybersecurity projects showcasing incident response, encryption, and threat analysis capabilities.

TECHNICAL SKILLS

Security: Penetration Testing, Cryptography (AES-256, RSA, HMAC), Incident Response, MITRE ATT&CK; Framework, Threat Analysis, Vulnerability Assessment, Network Security, Secure Coding

Languages: Python, JavaScript, Bash, SQL

Backend: Flask, FastAPI, Node.js, Express.js, RESTful API Design

Frontend: React, HTML/CSS, Responsive Design, Dark Theme UI/UX

Databases: SQLite, PostgreSQL, Data Modeling

ML/Data: scikit-learn, pandas, Feature Engineering, Classification Models, Cross-Validation

Tools & Platforms: Git, Kali Linux, Nmap, Burp Suite, Metasploit, Docker, DVWA, Metasploitable2

PROFESSIONAL EXPERIENCE

Cybersecurity Portfolio Developer (Freelance) | Nairobi, Kenya | 2024 - Present

Designed, developed, and deployed 11 production-grade cybersecurity projects including incident response simulators, phishing detection models, port scanners, and encryption tools. Implemented full-stack security applications using Flask, React, and cryptographic libraries. Created comprehensive documentation and GitHub repositories for each project.

Key Projects: SOC Incident Response Simulator (30+ MITRE ATT&CK; scenarios), Phishing Detection Model (94% precision, scikit-learn), Professional Port Scanner (parallel TCP scanning, CVE correlation), Hybrid File Encryption Tool (AES-256 + RSA), Network Enumeration Lab (Kali Linux reconnaissance)

AI/ML Evaluation Specialist (Freelance) | Outlier AI | 2024 - Present

Evaluated large language model outputs, performed quality assurance on AI system responses, and conducted prompt engineering for edge case analysis. Provided detailed feedback on model performance and areas for improvement.

EDUCATION

Bachelor of Science in Cybersecurity and Digital Forensics | Open University of Kenya | Expected 2028

Relevant Coursework: Threat Modeling (STRIDE, OAuth 2.0), Intrusion Detection Systems (Snort IDS/IPS), SQL Injection and Web Vulnerabilities (DVWA), Linux System Enumeration, Penetration Testing

FEATURED PROJECTS

SOC Incident Response Simulator

React + Node.js full-stack application with 30+ MITRE ATT&CK-tagged; incident scenarios, alert triage scoring engine, and leaderboard system for training SOC analysts on incident response workflows.

Phishing Email Detection Model

scikit-learn supervised classifier with 94% precision and 91% recall. Implements feature engineering for email headers, content analysis, and NLP signals. Includes cross-validation and feature importance analysis.

Professional Port Scanner

Flask backend with parallel TCP port scanning (50 concurrent threads). React UI with real-time progress, service detection, CVE correlation, and exportable results (JSON/CSV). Identifies 15+ common services.

Hybrid File Encryption Tool

Flask + React application implementing AES-256-GCM symmetric encryption and RSA hybrid cryptography. PBKDF2 key derivation (100K iterations), HMAC integrity verification, and secure key management UI.

Network Enumeration Lab

Hands-on reconnaissance against Metasploitable2 using Kali Linux. Demonstrates Nmap port scanning, enum4linux SMB enumeration, SNMP information leakage exploitation, and attack surface analysis.

CERTIFICATIONS & ACHIEVEMENTS

- Developed and deployed production-grade security applications on GitHub with 1000+ lines of code each
- Implemented cryptographic systems meeting NIST standards (PBKDF2, AES-256, RSA-2048+, HMAC-SHA256)
- Built machine learning classification models with cross-validation and performance optimization
- Conducted network reconnaissance and vulnerability assessment labs on intentionally vulnerable systems
- Created comprehensive technical documentation and professional portfolio site (GitHub Pages)